

```

1
2
3 -- CTE's allow using one or more SELECT queries to predefine temporary
4 -- tables that can be referenced.
5
6 -- CTE (Common Table Expression) Example
7 -- Determine how many counties in each state have a population of 100,000 or more.
8
9 WITH large_counties (county_name, state_name, pop_est_2019)
10
11 AS ( SELECT county_name, state_name, pop_est_2019
12 FROM us_counties_pop_est_2019
13 WHERE pop_est_2019 >= 100000)
14
15 SELECT state_name, count(*)
16 FROM large_counties
17 GROUP BY state_name
18 ORDER BY count(*) DESC
19 ;

```

Data Output Messages Notifications



Showing rows: 1 to 50

	state_name text	count bigint
1	Texas	40
2	Florida	36
3	California	35
4	Pennsylvania	31
5	New York	28
6	North Carolina	28
7	Ohio	27

```

1  WITH
2  counties (st, pop_est_2018) AS
3  (SELECT state_name, sum(pop_est_2018)
4  FROM us_counties_pop_est_2019
5  GROUP BY state_name),
6
7  establishments (st, establishment_count) AS
8  (SELECT st, sum(establishments) AS establishment_count
9  FROM cbp_naics_72_establishments
10 GROUP BY st)
11
12 SELECT counties.st,
13        establishment_count,
14        round((establishments.establishment_count /
15              counties.pop_est_2018::numeric(10,1)) * 1000, 1)
16        AS estabs_per_thousand
17 FROM counties JOIN establishments
18 ON counties.st = establishments.st
19 ORDER BY estabs_per_thousand DESC
20 ;
21

```

Data Output Messages Notifications









SQL

	st text	establishment_count bigint	estabs_per_thousand numeric
1	District of Columbia	2754	3.9
2	Montana	3569	3.4
3	Vermont	1991	3.2
4	Maine	4282	3.2
5	Wyoming	1808	3.1
6	Rhode Island	3152	3.0
7	Alaska	2186	3.0
8	New Hampshire	3841	2.8
9	Oregon	11796	2.8
10	South Dakota	2504	2.8
11	New York	55000	2.8

```
WITH us_median AS
    [(SELECT percentile_cont(.5)
    WITHIN GROUP (ORDER BY pop_est_2019) AS us_median_pop
    FROM us_counties_pop_est_2019)]

SELECT county_name,
    state_name AS st,
    pop_est_2019,
    us_median_pop,
    pop_est_2019 - us_median_pop AS diff_from_median
FROM us_counties_pop_est_2019 CROSS JOIN us_median
WHERE (pop_est_2019 - us_median_pop)
BETWEEN -1000 AND 1000
;
```

Data Output Messages Notifications

Showing rows 1 - 10					
	county_name text	st text	pop_est_2019 integer	us_median_pop double precision	diff_from_median double precision
	Cherokee County	Alabama	26196	25726	470
	Geneva County	Alabama	26271	25726	545
	Cleburne County	Arkansas	24919	25726	-807
	Johnson County	Arkansas	26578	25726	852
	St. Francis County	Arkansas	24994	25726	-732
	Montezuma County	Colorado	26183	25726	457
	Routt County	Colorado	25638	25726	-88
	Teller County	Colorado	25388	25726	-338
	Washington County	Florida	25473	25726	-253
	Butts County	Georgia	24936	25726	-790